Differentiating Other Functions

Exam code: 9MA0

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Help me understand

Differentiating other functions

How do I differentiate common functions?

- These are the common results
  - $\frac{d}{dx}(x^n) = nx^{n-1}$
  - $\frac{d}{dx}(e^x) = e^x$
  - $\frac{d}{dx}(a^x) = a^x \ln a$ for $a > 0$
  - $\frac{d}{dx}(\ln x) = \frac{1}{x}$
  - $\frac{d}{dx}(\sin x) = \cos x$
  - $\frac{d}{dx}(\cos x) = -\sin x$
  - $\frac{d}{dx}(\tan x) = \sec^2 x$
  - $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
  - $\frac{d}{dx}(\sec x) = \sec x \tan x$
  - $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

1. IF $y = e^{kx}$, WHERE k IS A REAL CONSTANT, THEN $\frac{dy}{dx} = ke^{kx}$

2. IF $y = \ln x$ THEN $\frac{dy}{dx} = \frac{1}{x}$

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3. IF $y = a^x$, WHERE a IS A REAL CONSTANT AND $a>0$, THEN $\frac{dy}{dx} = (\ln a)a^x$

4. IF $y = \ln kx$, WHERE k IS A REAL CONSTANT, THEN $\frac{dy}{dx} = \frac{1}{x}$

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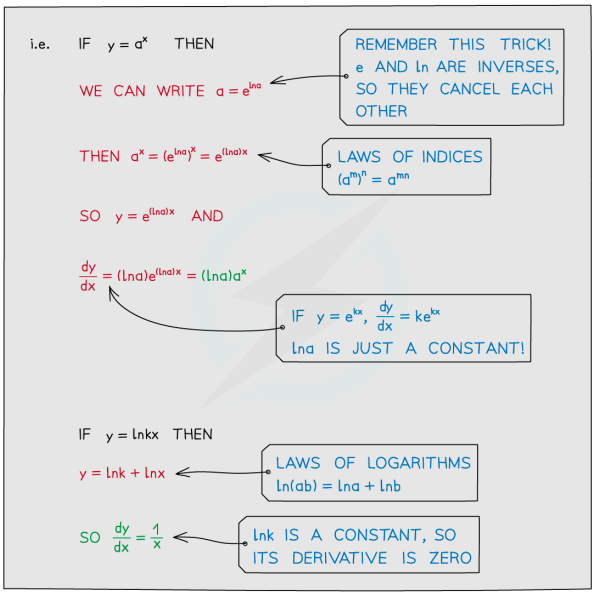

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- And for a^{kx}:

5. IF $y = a^{kx}$, WHERE a AND k ARE REAL CONSTANT AND $a>0$,
THEN $\frac{dy}{dx} = k(\ln a)a^{kx}$

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- This last formula can be derived from Formula 3 by using the **chain rule**

Examiner Tips and Tricks

- The formulae for some of these derivatives are not given in the formulae booklet – you need to know them.
- The formulae for $\mathbf{e^{kx}}$ and $\mathbf{\ln x}$ are the ones you absolutely do need to know.

, NOT $\frac{k}{x}$!

Worked Example

A curve has the equation $y = e^{-3x} + 3^{\frac{x}{2}} - \ln 5x$. Find the gradient of the curve at the point with x-coordinate 2.

$$\frac{dy}{dx} = -3e^{-3x} + \frac{1}{2}(\ln 3)3^{\frac{x}{2}} - \frac{1}{x}$$

FORMULA 1

FORMULA 4

FORMULA 5

WHEN $x = 2$,

$$\frac{dy}{dx} = -3e^{-6} + \frac{1}{2}(\ln 3)3^{1} - \frac{1}{2}$$

$$= \frac{1}{2}(3\ln 3 - 1) - 3e^{-6} \qquad (\approx 1.14)$$

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